MIL-S-15573A
10 APRIL 1952
SUPERSEDING
MIL-S-15573(SHIPS)
1 AUGUST 1950

MILITARY SPECIFICATION

STRAINERS, SEDIMENT, PIPELINE (FOR GASOLINE SERVICE)

This specification was approved by the Departments of the Army, the Navy, and the Air Force for use of procurement services of the respective Departments.

1. SCOPE

1.1 Scope.—This specification covers sediment strainers of the basket type for use in gasoline service lines for protection of meters.

1.2 Classification.—Strainers shall be of the following sizes, as specified (see 6.1):

1½ inches.
2 inches.
3 inches.
4 inches.
5 inches.
6 inches.
8 inches.

2. APPLICABLE SPECIFICATIONS, STANDARDS, DRAWINGS, AND PUBLICATIONS

2.1 The following specifications, standards, and drawings, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

NN-B-591 —Boxes, Fiberboard, Wood-Cleated (for Domestic Shipment).
NN-B-601 —Boxes, Wood-Cleated-Plywood, for Domestic Shipment.
NN-B-621 —Boxes, Wood, Nailed and Lock-Corner.

MILITARY

JAN-P-104 —Packaging and Packing for Overseas Shipment — Crates, Sheathed, Wood, Nailed.
JAN-P-105 —Packaging and Packing for Overseas Shipment—Boxes, Wood, Cleated, Plywood.
JAN-P-106 —Packaging and Packing for Overseas Shipment—Boxes, Wood, Nailed.
JAN-P-132 —Packaging and Packing for Overseas Shipment—Crates, Unsheathed, Wood; Nailed (for Maximum Net Load of 2,500 Pounds).
JAN-P-139 —Packaging and Packing for Overseas Shipment — Plywood, Container-Grade.
JAN-P-775 —Packaging and Packing for Overseas Shipment of Hose, Rubber or Fabric (Including Tubing); and Fittings, Nozzles, and Strainers.
MIL-B-857 —Bolts, Nuts, Studs, and Tap-Rivets (and Material for Same).
MIL-N-894 —Nickel - Copper - Alloy; Wrought.
MIL-D-963 —Drawings, Production (for Electrical and Mechanical Equipment for Naval Shipboard Use).
MIL-C-6057 —Crates and Boxes; Aircraft and Airframe Component Parts (General Specification for).
MIL-B-16541 —Bronze, Valve: Castings.

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MIL–M–16576—Metal, Gun: Castings.
MIL–W–16602—Wire, Spring, Phosphor-Bronze.

NAVY DEPARTMENT

General Specifications for Inspection of Material.

STANDARDS

MILITARY

MIL–STD–105—Sampling Procedures and Tables for Inspection by Attributes.
MIL–STD–129—Marking of Shipments.

DRAWINGS

BUREAU OF SHIPS

B–104 —Fittings, Pipe, Flanged Composition: Pressures, 100 Pounds and Below.
B–176 —Flanges, Composition, Silver Brazing, for Copper Nickel Alloy Tubing and Brazed Pipes, Pressures 100 Pounds and Below.

(Copies of specifications, standards, and drawings, required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

2.2 Other publications. — The following publication, of the issue in effect on date of invitation for bids, unless otherwise stated, forms a part of this specification:

NATIONAL BUREAU OF STANDARDS

Handbook H28—Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D. C.)

3. REQUIREMENTS

3.1 Materials.

3.1.1 *Cast parts.*—The body shall be made of gun metal conforming to Specification MIL–M–16576 or of bronze conforming to Specification MIL–B–16541.

3.1.2 *Wrought parts.*—All wrought parts shall be of material conforming to Specification MIL–N–894.

3.2 General design.—The strainers shall consist of a body with flanged inlet and outlet connections on a common axis, a removable strainer basket and cover plate, rigidly secured. Unless otherwise specified in the contract or order, the complete unit shall be designed for a working pressure of 100 p.s.i. and a hydrostatic test pressure of 200 p.s.i.

3.2.1 *Body.*—The body shall be designed to receive the strainer basket in such position to permit flow through the basket from the inside out. The inlet and outlet shall be flanged and located on a common axis through the strainer body. Provision shall be made for rigidly retaining the strainer basket in the body. A boss shall be provided on the bottom of the strainer well to permit drilling for attachment of a 1-inch drain line on installation. An arrow indicating the direction of flow shall be cast on the body.

3.2.2 *Strainer basket.*—The strainer basket shall be of nickel-copper alloy wire cloth, and shall be of double wall design, consisting of an 80-mesh basket within a shroud basket of wire mesh or perforated sheet metal to prevent breakdown of the inner screen under operating conditions. The effective clear area through the complete strainer basket and shroud shall be not less than 10 times the area of the service line, the inside diameter of which should be taken as the nominal size. The complete element shall be separable and provided with lifting bale. The strainer basket assembly shall have seal surfaces on the mating faces of basket collars and body supports to prevent by-passing of any fluid through the baskets. Means shall be provided for locking the inner screen to the shroud to eliminate chatter. The assembly of strainer baskets shall be accomplished by welding or silver brazing. Solder will not be accepted.

3.2.3 *Cover.* — Unless otherwise specified in the contract or order, the strainer well

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shall be provided with a cover secured by bolts or studs and nuts. Dog clamps or drop bolts, when specified (see 6.1) for clamping the cover, shall be of adequate strength and designed to permit ready removal of the cover plate.

3.2.4 Bolts or studs and nuts shall be of nickel-copper alloy, and shall conform with Specification MIL–B–857. Studs if used shall have a No. 5 thread fit on the set end, No. 3 thread fit on nut end. Threads shall conform to Handbook H28.

3.2.5 *Gaskets.*—Gaskets shall be of a design and material suitable for the intended use, consideration being given to its resistance to gasoline, oil and sea water and to the tightness of seal. Cork impregnated Buna-N, or spiral wound metal-teflon, or equivalent, will be generally acceptable.

3.2.6 *Springs.*—Springs, if used, shall be of phosphor bronze conforming to Specification MIL–W–16602.

3.2.7 *Flanges.*—Flanges of the inlet and outlet connections shall conform to the dimensions shown on Drawing B–104 with drilling in accordance with Drawing B–176. All flanges shall be finished with a phonographic finish on gasket seat areas. The inlet and outlet cover flanges also shall be machined with a phonographic finish with 60 to 80 cuts per inch of gasket face width, 0.002 to 0.003 inch deep cut at an average speed of 50 feet per minute.

3.2.8 *Operation.*—All strainers shall be so designed that the pressure drop through the unit with clean fuel does not exceed 2 p.s.i. maximum, at the following rated capacity:

Size Inches	*Minimum capacity g.p.m.*
1¼	125
2	125
3	350
4	650
5	800
6	1,000
8	1,500

3.3 Repair parts.—Repair parts shall be furnished as follows: (One set = 100 percent per strainer.)

Gaskets, one set for each strainer.
Springs, one set for each strainer.
Strainer baskets, one set for each strainer.

3.4 Special tools.—Special tools for servicing the strainer, if required, shall be furnished.

3.5 Drawings.—Prior to production, the contractor shall furnish drawings of the strainer for the approval of the bureau or agency concerned. Drawings shall conform to type I of Specification MIL–D–963, and shall show the details of design, and the size, capacity, and type of connection.

3.6 Workmanship.—Strainers shall be free from defects which affect operation or appearance. Castings shall be clean, sound, and free from blow holes, hard spots, porosity, cracks and other injurious defects. They shall be smooth and well cleaned both inside and outside. All fins and roughness shall be removed. Castings shall not be repaired, plugged, impregnated, brazed, or burned. The workmanship shall be first class in every respect.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Inspection procedures.—For Naval purchases, the general inspection procedures shall be in accordance with General Specifications for Inspection of Material.

4.2 Inspection lot.—All strainers of the same size presented at one time shall be considered a lot for purposes of acceptance inspection and tests.

4.3 Sampling for lot acceptance.

4.3.1 *Sampling for inspection.*

4.3.1.1 *Sampling for intermittent procurement of isolated lots.*—A random sample of

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strainers shall be selected from each inspection lot of material offered for Government inspection of visual and dimensional characteristics with lot acceptance based on the single sampling inspection requirements of table I, in accordance with Standard MIL–STD–105.

TABLE I.—*Sampling for visual and dimensional inspection AQL (approx.) = 1.5 percent defective*

Number of strainers in inspection lot	Number of strainers in sample	Acceptance number (defectives)	Rejection number (defectives)
15 and under	7	0	1
16 to 40	10	0	1
41 to 110	15	0	1
111 to 300	25	1	2
301 to 500	35	1	2
501 to 800	50	2	3
801 to 1300	75	3	4
1301 and over	110	4	5

4.3.1.2 *Sampling for continuous production.*—Where there is continuous production for the Government and inspection lots include at least 200 items, sampling for Government inspection shall be conducted in accordance with Standard MIL–STD–105 at Inspection Level I and AQL = 1.5 percent defective, the type of sampling to be determined by the Government inspector.

4.3.2 *Sampling for lot acceptance tests.*—A random sample of strainers shall be selected from each inspection lot in accordance with table II, and shall be subjected to the lot acceptance tests specified in 4.5.

4.3.3 *Test coupons.* — Two test coupons shall be taken from each heat of body castings for chemical analysis as specified in 4.7.

4.4 **Inspection.**

4.4.1 *Visual and dimensional inspection.*—Each of the sample strainers selected in accordance with table I shall be visually and dimensionally inspected by the Government inspector to verify compliance with this specification. Any strainer in the sample containing one or more visual or dimensional defects shall be rejected, and if the number of defective strainers in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

TABLE II.—*Sampling for proof and service tests AQL (approx.) = 1.0 percent defective*

Number of strainers in inspection lot	Number of strainers in sample	Acceptance number (defectives)	Rejection number (defectives)
15 and under	5	0	1
16 to 40	7	0	1
41 to 110	10	0	1
111 to 300	15	0	1
301 to 500	25	0	1
501 to 800	35	1	2
801 to 1300	50	1	2
1301 and over	75	2	3

4.5 **Lot acceptance tests.**—Each sample strainer selected in accordance with 4.3.2 shall be tested as specified in 4.5.1 and 4.5.2 to verify conformance with this specification. Any strainer in the sample failing one or more tests shall be rejected, and if the number of defective strainers in the sample exceeds the acceptance number for table II, the lot represented by the sample shall be rejected.

4.5.1 Each sample strainer shall be tested hydrostatically with kerosene or gasoline for strength and porosity to a pressure of 200 p.s.i. The pressure shall be maintained for not less than 15 minutes. Leakage, sweating or visible deformation at any point on the surface shall be cause for rejection.

4.5.2 Each sample strainer shall be tested under water with 100 p.s.i. air for porosity of body. Leakage under this test shall be cause for rejection.

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4.6 Rejected lots.—Rejected lots may be offered again for Government inspection provided the contractor has removed all nonconforming strainers. The Government inspector shall select and examine samples from such resubmitted lots to verify compliance with this specification.

4.7 Chemical analysis.—The sample selected in accordance with 4.3.3 shall be analyzed to determine conformance to the applicable material specification. Failure to meet the chemical requirements shall be cause for rejection of the lot.

5. PREPARATION FOR DELIVERY

5.1 Preservation.

5.1.1 *For domestic shipment.*—The application of rust-preventive compounds beyond those normally applied to strainers in best commercial practice will not be required.

5.1.2 *For oversea shipment.* — Strainers shall be perserved as specified in Specification JAN–P–775.

5.2 Packing.

5.2.1 *For domestic shipment.*

5.2.1.1 Strainers weighing 200 pounds or less shall be packed in wood-cleated fiberboard, wood-cleated plywood, or nailed wood boxes conforming to Specification NN–B–591, NN–B–601, or NN–B–621, respectively.

5.2.1.2 Strainers weighing in excess of 200 pounds shall be packed in nailed wood, sheathed nailed wood, or unsheathed nailed wood crates conforming to Specification MIL–C–6057, JAN–P–104, or JAN–P–132, respectively. The strainers shall be secured by adequate blocking and bracing.

5.2.2 *For oversea shipment.*

5.2.2.1 Strainers weighing 200 pounds or less shall be packed in cleated plywood boxes or nailed wood boxes conforming to Specification JAN–P–105 or JAN–P–106, respectively. Plywood shall conform to type A or B, condition I of Specification JAN–P–139.

5.2.2.2 Strainers weighing in excess of 200 pounds shall be blocked and braced in nailed wood crates conforming to Specification MIL–C–6057, JAN–P–104 or JAN–P–132, respectively.

5.2.3 *Strapping.* — Shipping containers shall be strapped in accordance with the applicable container specification.

5.3 Marking.—In addition to any special marking required by the contract or order, shipments shall be marked in accordance with Standard MIL–STD–129.

6. NOTES

6.1 Ordering data. — Procurement documents should specify the following:

(a) Title, number and date of this specification.

(b) Size (inlet and outlet) and capacity required (see 1.2 and 3.2.8).

(c) Whether dog clamps or drop bolts are required in lieu of bolts or studs (see 3.2.3).

Notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

Custodians:
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Army—Corps of Engineers

Other interest:
Navy—S.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD, FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE C-6378

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